

Ryu Hyunseung

I build web apps — and the languages and runtimes they run on.

I move between frontend work and systems programming. Using an abstraction layer and building one teach different things, and I want both.

South Korea ryu@rhseung.me github.com/rhseung rhseung.me

Experience

Audio Intelligence Technology & Research Lab (AiTeR), School of EECS, GIST

2025.09 – Present

Frontend developer intern

Taking part in experiments and research at the audio intelligence lab.

aiter.gist.ac.kr/ ↗

Education

Gwangju Institute of Science and Technology (GIST)

2025.03 – Present

B.S., School of EECS and Department of AI Convergence (double major)

First year is undeclared; majors are chosen from the second year.

Freshman representative, 2025 matriculation ceremony

Academic excellence scholarship, spring 2025

GS1001 Calculus and Applications — 1st, 240 / 240

GS1401 Computer Programming — 1st, 98 / 100

Gwangju Science Academy for the Gifted

2022.03 – 2025.02

Natural Sciences

Computer science GPA 4.285 / 4.3

Gold Prize, 2023 in-school competition (computer science)

Gold Prize, 2022 in-school competition — the only one awarded to a first-year student

Subject Excellence Award, Computer Science Project — 1st, year 3 semester 1

Data Structures & Algorithms — 2nd, year 3 semester 1

Subject Excellence Award, Programming & Problem Solving — 1st, year 2 semester 2 (AP)

Subject Excellence Award, Object-Oriented Programming — 1st, year 2 semester 2

Subject Excellence Award, Machine Learning & Deep Learning — 1st, year 2 semester 2

Subject Excellence Award, Computer Science II — 2nd, year 1 semester 2

Subject Excellence Award, Computer Science I — 5th, year 1 semester 1

Selected projects

rhseung.me	2026.08
A personal site built on Astro islands. This page is the result.	
Compass	2024.11 – 2024.11
A companion service that gathers everything you need for campus life in one place.	
Table-tennis robot	2024.10 – 2025.01
Computer vision tracks the ball in real time and predicts the strike point to drive a four-axis linear robot.	
NEAT	2024.09
Evolving network topology itself with NEAT (NeuroEvolution of Augmenting Topologies).	
Glance	2024.06
A Minecraft interface mod bundling a set of quality-of-life tweaks.	
Fliggle	2024.05 – 2024.07
A social app built for practice, with both the frontend and the backend written from scratch.	
IdP	2024.03 – 2024.08
GIST's single sign-on system. OAuth 2.0 and OpenID Connect let one account log into every campus service, and external ones too.	
Particles	2024.03 – 2024.11
A constraint-based 2D physics engine. Position-based dynamics solve rigid bodies and joints.	
Pattern	2024.02 – 2024.05
Builds regular expressions through a builder API — method chains instead of unreadable expressions.	
DateTime	2024.01 – 2024.04
Parses natural-language expressions like "next Tuesday" into dates.	
siunits	2023.12
A unit-arithmetic library for physics experiments. Dimensionally invalid operations fail at the type level.	
GSA KakaoTalk bot	2023.09 – 2024.02
A KakaoTalk bot for Gwangju Science Academy — meals, timetables and the academic calendar without leaving the chat.	
YOLOv3 from scratch	2023.03 – 2023.11
Reimplemented YOLOv3 from the paper alone — anchor boxes and multi-scale detection included.	

Competitions

3rd place, 2026 AX Challenge Track 1	2026.08
Table-tennis robot track · Jun – Aug 2026	

Winner, 2025 Honam Regional Mini Tex-Corps Ministry of Science and ICT · COMPA Hosted by GIST, Ewha Womans University and Wonkwang University startup centers	2025
Participant, AI Creative Convergence Competition GIST AI Graduate School Table-tennis robot track	2025
Participant, JUNCTION ASIA 2025 ENTBE	2025
Participant, L:AUNCH AI Career School Root Impact · Krypton X Supported by Google.org	2025

Skills

Languages

- TypeScript
- JavaScript
- Python
- Java
- Kotlin
- C
- C++
- Dart

Web

- React
- Next.js
- Astro
- TanStack Router
- TanStack Query
- React Router
- Tailwind CSS
- Vanilla Extract
- Styled Components
- i18next
- Vite
- HTML
- CSS

Backend & data

- Node.js
- Nest.js
- Flask
- Prisma
- MySQL
- OpenAPI TypeScript

Machine learning

- PyTorch
- TensorFlow
- Keras
- scikit-learn
- OpenCV
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Altair
- Jupyter

Apps & graphics

- Flutter
- SFML
- Swing

Tooling

- Bun
- npm
- Yarn
- PyPI
- Figma
- LaTeX